4.1 What Is an Integral?

WORKSHEET

The question

STEP III 1989, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q10

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The two sums in part (iii) are sums of areas of rectangles of width $\frac{a}{n}$. For an increasing function f, which sum is an underestimate of the area under the curve, and which an overestimate?

STAGE 1 A sum that telescopes

Covers: Part (i)

Ask yourself Let $P(r) = r(r+1)(r+2)(r+3)(r+4)(r+5)$, a product of six consecutive integers. Work out $P(r) - P(r-1)$: which factors do they share?

Pause and try Add your result from $r = 1$ to n. Most terms cancel. Then compare r^5 with $r(r+1)(r+2)(r+3)(r+4)$.

STAGE 2 A lower bound

Covers: Part (ii)

Pause and try Do the same with a product of consecutive integers going *down*, ending at r: $(r-4)(r-3)(r-2)(r-1)r$. Add from $r = 0$ to $n-1$, and compare each term with r^5. Why is the inequality strict?

STAGE 3 The integral

Covers: Part (iii)

Ask yourself For an increasing function, which of the two sums is an underestimate of the area, and which an overestimate?

Pause and try Write both sums for $f(x) = x^5$ in terms of $\sum r^5$, and use parts (i) and (ii). What do the bounds tend to as $n \to \infty$?

What's really going on

Riemann's definition

The idea of finding areas by summing thin strips goes back to Archimedes, who found the area under a parabola by a version of exactly this argument around 250 BC. Newton and Leibniz turned integration into the reverse of differentiation, and for a century and a half that was how most people thought about it. In 1854 Bernhard Riemann gave the definition that university courses still begin with: divide $[a, b]$ into small pieces, form sums of (width) $\times$ (height of f somewhere in the piece), and say f is integrable if all such sums approach one limit as the pieces shrink. The version in this question, with equal strips and left and right endpoints, is a special case that works for increasing functions; Jean-Gaston Darboux's "upper and lower sums" make the general case clean.

Not every function has an integral

With a precise definition, you can ask which functions are integrable. Every continuous function on a closed interval is, and every increasing function. But consider Dirichlet's function: $f(x) = 1$ if x is rational and 0 if irrational. Every strip, however thin, contains both rationals and irrationals, so every upper sum is 1 and every lower sum is 0. They never agree, and $\int_0^1 f$ does not exist in Riemann's sense. In 1902 Henri Lebesgue invented a more powerful integral that handles this function (its Lebesgue integral is 0) and behaves much better under limits. The Lebesgue integral is the foundation of modern probability theory, and you will meet it in your second or third year.

Discrete calculus

The telescoping in Stage 1 is calculus in disguise. The "rising factorial" $r(r+1)\cdots(r+k-1)$ behaves under summation exactly as x^k behaves under integration: summing it raises the number of factors by one and divides by the new number of factors, just as $\int x^k\,\mathrm{d}x = \frac{x^{k+1}}{k+1}$. This parallel between sums and integrals, differences and derivatives, is called the *calculus of finite differences*, and it is a powerful tool for evaluating sums.

Try this next

1. **A limit as an integral.** Try STEP III 1999, Question 3 (step.maths.org/questions/99-s3-q3). It turns limits of sums into integrals: in each sum, look for rectangles under a curve.

2. **Archimedes' parabola.** Using $\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$ and the definition in part (iii), prove that $\int_0^a x^2\,\mathrm{d}x = \frac{1}{3}a^3$.

3. **Rising factorials.** Show that $\displaystyle\sum_{r=1}^{n} r(r+1)\cdots(r+k-1) = \frac{n(n+1)\cdots(n+k)}{k+1}$ for every positive integer k.

4. **The exact sum.** Use the identity $r(r+1)(r+2)(r+3)(r+4) = r^5 + 10r^4 + 35r^3 + 50r^2 + 24r$ and the known formulas for $\sum r^k$ with $k \le 4$ to find $\sum_{r=1}^{n} r^5$ exactly.

Further reading

- **David Bressoud, *A Radical Approach to Lebesgue's Theory of Integration*** (Cambridge University Press). The story from Riemann's integral to Lebesgue's, told through the problems that forced each step.
- **Michael Spivak, *Calculus*** (Cambridge University Press). Chapter 13 defines the integral by upper and lower sums, exactly as here.
- **Graham, Knuth & Patashnik, *Concrete Mathematics*** (Addison-Wesley). Chapter 2 develops the calculus of finite differences, with rising and falling factorials.

The full guided solution is in Section 4.1:
https://beyondstep.keithmcnulty.org/ch04/riemann.html